Case Studies

Behavioural Stress Index — applied case-by-case

A reference document — not for live presentation. Each slide documents one case the framework was applied to: what fired, what was suppressed, what the gates said, what happened.

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FIN 580 · Spring 2026 · github.com/vermasidd1502/bnpl-trap

All cases at a glance

Nine firms across BNPL, subprime auto, and marketplace lending — plus one architectural case study.

CASE	DATE	VERDICT	WHAT BSI DID	OUTCOME
Carvana 2022	Sep 2021	TRUE POSITIVE	weekly z=2.65 fired six weeks after $370 peak	−98% in 15 mo
Bridgecrest 2022	2022-08-15	TRUE POSITIVE (formal)	Carvana subsidiary; co-event detection	co-event
Upstart 2022	2022-05-09	TRUE POSITIVE (formal)	detected ex-ante; warehouse-line earnings	−85% peak-to-trough
CURO 2024	2024-03-25	EXCLUDED (ingest)	detected; SEC name change broke ingest	Chapter 11
Affirm 2023	May 2023	TRUE POSITIVE	z=2.30 fired 18 days before Q3 guidance cut	−27% / 2 sessions
Klarna 2024	Sep 2024	OBSERVED	BSI fired; framing depends on interpretation	−15% / 3 sessions
Sezzle 2024	Sep 2024	FALSE POS · SUPPRESSED	normalisation gate caught it ex-ante	trade not routed
LendingClub 2025	Apr 2025	TRUE POSITIVE	fired 11 days before NCO step	−18% / 4 sessions
Tricolor 2025	2025-09-10	TRUE POSITIVE (formal)	detected; appeared in sensitivity_full.csv	Chapter 7
Two-pod architecture	–	ARCHITECTURAL	BearWatch + Equity Monitor isolation	no double-count

Carvana 2022 — the canonical case

BSI fired six weeks after the all-time high. Captured 98% drawdown over fifteen months.

TRUE POSITIVE · 98% drawdown captured

KEY FACTS

SUB-SECTOR	**Subprime auto / used-car retail**
FIRST BSI FIRE	**Sep 20, 2021 · weekly z = 2.65**
STOCK AT FIRE	**$327 pre-split (−12% from peak)**
PEAK	**Aug 10, 2021 · $370 (pre-split)**
STRONGEST FIRE	**Apr 11, 2022 · z = 3.01 · $106**
TROUGH	**Dec 30, 2022 · $4.74 · −99% peak**

NARRATIVE

Carvana sat at the intersection of two distress vectors: the COVID-era used-car bubble unwind (sector cyclical) and subprime-auto credit deterioration (firm-specific). The first vector took the stock from $370 to roughly $140 by early 2022 — sector noise, not BSI's job to catch.

BSI's contribution starts in September 2021. CFPB complaint volume began accelerating six weeks after the peak. By April 2022, BSI had been firing intermittently for seven months, with the strongest signal (z = 3.01) marking the secondary leg lower.

LESSON

BSI is a firm-specific distress detector. It stays silent during macro / sector cycles but fires decisively when consumer-side data starts deteriorating ahead of the equity / credit markets.

Affirm 2023 — true positive, fast cycle

BSI z = 2.30 fired 18 days before the Q3 fiscal-year guidance cut. Stock fell 27% in two days post-print.

TRUE POSITIVE · all 5 gates fired

KEY FACTS

SUB-SECTOR	**BNPL / fintech consumer lending**
BSI FIRE	**May 11, 2023 · z = 2.30**
LEAD TIME	**18 days before guidance cut**
CATALYST	**May 9, 2023 earnings call**
DRAWDOWN	**−27% in 2 trading sessions**
ALL GATES FIRED?	**YES — 5 of 5 passed**

NARRATIVE

Affirm's fiscal Q3 2023 (reported May 9) showed sharp deterioration in transaction-volume growth and provision-for-credit-losses ratio. Market reaction was a 27% one-day drop.

BSI was elevated for ~18 days before the print. The signal was driven by CFPB complaint acceleration plus app-store sentiment — consumer-side, with no macro or CCD divergence. The gates correctly classified it as firm-specific stress, not a broad consumer-credit selloff.

LESSON

When all five gates fire AND the market catalyst follows within the trade window (T+3), the cascade thesis is validated end-to-end on a single name.

Sezzle 2024 — false positive, suppressed

Headline BSI fired at z = 2.60 in Q3 2024. The denominator-normalisation gate caught it ex-ante — no trade routed.

FALSE POSITIVE · normalisation gate suppressed

KEY FACTS

SUB-SECTOR	**BNPL / consumer-pay-in-4**
BSI FIRE	**Sep 22, 2024 · z = 2.60**
SUPPRESSION GATE	**Denominator-normalisation (paper §7.3)**
WHY SUPPRESSED	**Complaint-per-customer rate FLAT**
UNDERLYING CAUSE	**Volume spike was a growth artefact**
VERIFICATION	**No fundamental deterioration in 90d**

NARRATIVE

Sezzle was scaling its user base aggressively in 2024. Raw complaint volume rose because the customer base grew — not because per-customer satisfaction was deteriorating.

The denominator-normalisation gate (paper §7.3) is built for exactly this scenario. It divides complaint volume by an estimate of active customers, then re-z-scores. The normalised z stayed below threshold even as the raw z hit 2.60.

Klarna 2024 had a similar pattern around its IPO down-round (z = 2.60 on Sep 5). The

LESSON

Same headline z. Different downstream gate behaviour. The normalisation gate separates real consumer-pain spikes from volume-driven growth artefacts. This is the canonical false-positive-caught story for the framework.

LendingClub 2025 — fast-cycle true positive

BSI z = 2.45 fired 11 days before the NCO ratio step in the Q1 2025 print. Stock fell 18% in 4 sessions.

TRUE POSITIVE · shortest lead in the panel (11 days)

KEY FACTS

SUB-SECTOR	**Marketplace lending / consumer**
BSI FIRE	**Apr 19, 2025 · z = 2.45**
CATALYST	**Apr 30, 2025 — Q1 NCO ratio step**
LEAD TIME	**11 days (shortest in narrative cases)**
DRAWDOWN	**−18% in 4 trading sessions**
PILLARS DRIVING	**CFPB + EDGAR XBRL fundamentals**

NARRATIVE

LendingClub is the cleanest 2025 case. The Q1 print revealed a step-change in net charge-offs — not a smooth ramp, a step. That kind of discontinuity is exactly what the cascade thesis predicts: hard accounting metrics deteriorating after the soft signals have already moved.

BSI fired at z = 2.45 eleven days ahead of the print. That's the shortest lead time in the five narrative cases — but importantly, it was already accelerating in March, with weekly z values climbing from 1.6 → 1.9 → 2.45 over three weeks.

complaint data.

LESSON

Marketplace-lending firms have similar cascade dynamics to BNPL — the framework generalises. The G5 FDS gate is essential when the lead time is tight; it reads the upstream accounting before the equity catches up.

Tricolor 2025 — formal EVENT firm, terminal-stage detection

Detected as one of the 5/5 events in sensitivity_full.csv. Filed Chapter 7 bankruptcy in September 2025.

TRUE POSITIVE (FORMAL) · first BSI-detected outright bankruptcy

KEY FACTS

SUB-SECTOR	**Subprime auto (deep tier)**
EVENT DATE	**Sep 10, 2025 · Chapter 7 filing**
SENSITIVITY ROW	**DETECTED in sensitivity_full.csv**
LEAD TIME	**approx. 5 months prior — first fire**
LEGAL ENTITY	**Tag Intermediate Holding Company, LLC**
NOTABLE	**First terminal event in the panel**

NARRATIVE

Tricolor is a deep-subprime auto lender — sub-FICO 550 borrowers, low-income consumers, often immigrant communities. Their 2025 Chapter 7 filing is the first terminal event the panel captured.

Methodologically important: Tricolor is a private company. There is no public equity to reference, no traditional analyst coverage, no Bloomberg sell-side notes. BSI detected the distress purely from CFPB complaint trajectory plus the firm's vitality measure.

This case extends the framework's reach: BSI is not just a public-equity tool. It works on

LESSON

The framework generalises beyond public equities. Any non-bank consumer lender that has CFPB complaint history is reachable. Terminal-stage detection (Chapter 7) is feasible months ahead with adequate CFPB coverage.

Upstart 2022 — formal EVENT firm, long-lead detection

Detected approximately 700+ days before the May 2022 warehouse-line earnings disaster.

TRUE POSITIVE (FORMAL) · long-lead detection

KEY FACTS

SUB-SECTOR	**Personal-loan / fintech lending**
EVENT DATE	**May 9, 2022 (Q1 earnings call)**
CATALYST	**Warehouse-line guidance disaster**
PRE-EVENT PRICE	**approx. $120 (mid-Apr 2022)**
POST-EVENT	**approx. $30 within 6 months**
SENSITIVITY	**DETECTED — fire approx. 700+ days prior**

NARRATIVE

Upstart is a credit-decisioning fintech that uses ML to underwrite personal loans, then funds them through a warehouse-line model. When the warehouse facilities tightened in early 2022, Upstart's funding capacity collapsed almost overnight — that was the May 2022 earnings event.

BSI's detection lead was unusually long: the first fire happened more than two years before the Q1 2022 print. Most of those early fires were below z = 3.0 — quieter signals — but the trend was clear in retrospect.

LESSON

Long lead-time detection is possible when the cascade is slow-moving. Upstart's warehouse-line failure was visible in CFPB complaint patterns years before it crystallised in equity. CURO is a reminder that schema/name-change drift is a real ingest-pipeline risk.

the main deck.

Two-pod architecture — BearWatch + Equity Monitor isolation

Architectural case study, not a firm. Demonstrates why two independent decision logics on the same ticker are methodologically ~~clean~~er than one fused pod.

ARCHITECTURAL · multi-strategy isolation by design

BEARWATCH POD

Alt-data signal layer

Inputs: CFPB complaints, App Store, Reddit, Bluesky, Google Trends, ABS, macro confound, vitality.

Decision: 5-gate stack (BSI · SCP · MOVE · CCD · FDS).

Trades: bearwatch journal · own position cap.

Use case: consumer-credit distress detection.

EQUITY MONITOR POD

Technical / macro layer

Inputs: Wozniak / CMT — Fibonacci, U/D volume, OBV, RSI, Stochastic, Bollinger, ATR.

Decision: 5-archetype debate (Renaissance · Bridgewater · Two Sigma · Millennium · Citadel).

Trades: equity-monitor journal · own position cap.

Use case: technical / macro signals on any ticker (open universe).

CROSS-SOURCE EXPOSURE CAP

Joint-portfolio safeguard. Reads BOTH journals; if combined open exposure on a single ticker would exceed 20% of equity, the operator gets a warning and must explicitly override. Prevents either pod from independently breaching the joint position limit.

How to verify any case in this deck

Every claim above is queryable against the warehouse and result CSVs in the GitHub repo and Dropbox bundle.

VERIFY IN THE WAREHOUSE

```
import duckdb

con = duckdb.connect(
    'data/warehouse/warehouse.duckdb',
    read_only=True,
)

# CFPB volume per firm — weekly
df = con.execute('''
    SELECT
        date_trunc('week', received_at) AS wk,
        COUNT(*) AS n
    FROM cfpb_complaints
```

```
# Compute weekly z-score against
# 26-week rolling baseline
# (full code in scripts/run_all_ingest.sh)
```

AUTHORITATIVE RESULT FILES

paper_results/sensitivity_full.csv
5/5 events caught — DETECTED flags

paper_results/granger_aggregate.csv
23/27 firms reject H0

paper_results/bsi_panel.csv
panel headline beta = -0.082

paper_results/robustness.csv
alt-threshold sensitivity

paper_results/baseline_comparison.csv
macro vs CFPB vs BSI specs

paper_results/universe_27.csv
EVENT vs CONTROL classifications

bloomberg_exports/_combined_AFRMT_prices.csv

GITHUB	**github.com/vermasidd1502/bnpl-trap**
DROPBOX	830 MB bundle · paper + deck + data + pod · link in submission email